

XIAOHUI ZHANG

<https://tjzxh.github.io/academic-life/>

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EDUCATION

Ph.D., Traffic Engineering	2021 – 2025
Tongji University & National University of Singapore	China & Singapore
B.S. & M.S., Traffic Engineering	2013 – 2020
Tongji University	Shanghai, China
B.S., Mathematics with Applied Mathematics (Minor)	2013 – 2015
Tongji University	Shanghai, China

WORK EXPERIENCE

Alibaba Group – Amap (Gaode) – HD (High Definition) Map	2020 – 2021
<i>Data Engineer</i>	Beijing, China

PROFESSIONAL SKILLS

- Experienced in Python, MATLAB
- Experienced in traffic simulation software Vissim, SUMO
- Experienced in mathematical modeling and applying new AI methods

AWARDS & HONORS

▪ Outstanding Thesis Award	2020
▪ Scholarships of Tongji University	2014, 2015, 2019
▪ The Second Prize in the National Post-Graduate Mathematical Contest in Modeling	2017
▪ The Meritorious Winner in the Mathematical Contest in Modeling	2016
▪ The Second Prize in the Shanghai Mathematics Competitions	2014

RESEARCH INTERESTS

1. New AI technologies for autonomous driving

Overview: Apply and advance AI, including reinforcement learning and deep learning, to learn decision-making for autonomous driving.

- **Driving decision making with deep reinforcement learning.** DRL is applied in traffic simulator Vissim to learn the optimal driving policy, and further enhanced through a hierarchical framework integrating rule-based models, demonstrating both interpretability and adaptability in traffic flow optimization.
- **Learning human driving behaviors with deep learning.** Combining LSTM with data aggregation, the HRC-LSTM is proposed to minimize the cumulative error and precisely predict vehicle's trajectories, reproducing consistent car-following and lane-changing behaviors with empirical data.

Future Directions: Explore generative AI for traffic simulation and autonomous driving testing, and integrate AI methods with analytical modeling to enhance interpretability and trustworthiness.

2. Autonomous driving's impacts on traffic

Overview: Explore how the AI-driven AV reshapes traffic dynamics, with the goal of ensuring that AV serves the public good by enhancing, rather than disrupting, the overall traffic flow.

- **Stochastic foundations of traffic flow: From microscopic interactions to macroscopic dynamics.** Drawing on statistical physics, traffic is modeled as a complex stochastic system where microscopic leader–follower interactions are represented by conditional distributions. Applying Markov chain modeling, these interactions yield collective dynamics such as the flow–density relation and phase transition of capacity drop at critical density.
- **String stability analysis of neural network-based car-following models.** A theoretical framework consisting of equilibrium states estimation, non-differentiable function approximation, stability criteria derivation and partial derivative calculation is proposed to investigate the string stability of neural network based car-following models.
- **Traffic-level impacts of mixed traffic consisting of automated vehicles.** A data-driven probabilistic modeling framework is proposed for mixed traffic flow, enabling the input of real-world AV trajectory datasets and output of the stochastic flow–density relation under given AV penetration rates and AV spatial distributions.

Future Directions: Conduct holistic evaluations of AV impacts beyond efficiency, including safety, robustness; incorporate human factors (e.g., the degradation of AV and driver takeover); and develop traffic-adaptive control strategies to smooth the traffic.

PUBLICATIONS

With over **460 citations**, I have published multiple papers in leading journals in the field of transportation systems, including **Transportation Research Part C** and **Communications in Transportation Research**.

- (Under Review) **Zhang, X.**, Sun, Jie, Sun, Jian, 2025. A micro-macroscopic model of capacity drop: effects of disturbance propagation. *Transportation Research Part B: Methodological*. (**JCR Q1, IF 6.3**)
- **Zhang, X.**, Yang, K., Sun, Jie, Sun, Jian, 2025. Stochastic fundamental diagram modeling of mixed traffic flow: A data-driven approach. *Transportation Research Part C: Emerging Technologies* 179, 105279. (**JCR Q1, IF 7.9**)
- **Zhang, X.**, Sun, Jie, Sun, Jian, 2025. On the stochastic fundamental diagram: A general micro-macroscopic traffic flow modeling framework. *Communications in Transportation Research* 5, 100163. (**JCR Q1, IF 14.5**)
- **Zhang, X.**, Sun, Jie, Zheng, Z., Sun, Jian, 2024. On the string stability of neural network-based car-following models: A generic analysis framework. *Transportation Research Part C: Emerging Technologies* 160, 104525. (**JCR Q1, IF 7.9**)
- **Zhang, X.**, Sun, Jie, Wang, Y., Sun, Jian, 2023. A Hierarchical Framework for Multi-Lane Autonomous Driving Based on Reinforcement Learning. *IEEE Open Journal of Intelligent Transportation Systems* 4, 626–638. (**JCR Q1, IF 5.3**)
- **Zhang, X.**, Sun, Jie, Qi, X., Sun, Jian, 2019. Simultaneous modeling of car-following and lane-changing behaviors using deep learning. *Transportation Research Part C: Emerging Technologies* 104, 287–304. (**JCR Q1, IF 7.9**)
- Ye, Y.*, **Zhang, X.***, Sun, J., 2019. Automated vehicle's behavior decision making using deep reinforcement learning and high-fidelity simulation environment. *Transportation Research Part C: Emerging Technologies* 107, 155–170. (**JCR Q1, IF 7.9**)

ACADEMIC SERVICE & MENTORSHIP

- Reviewer for Transportmetrica A: Transport Science (JCR Q2, IF 3.1)
- Mentored graduate thesis project on mixed traffic flow modeling

LANGUAGE PROFICIENCY

- IELTS 7 (Listening 7 Reading 8.5 Writing 6.5 Speaking 6)

RELEVANT COURSES

- Mathematical Foundation of Reinforcement Learning, Machine Learning, Optimization Theory, Mathematical Modeling, Statistical Analysis, Mathematical Analysis
- Traffic Flow Theory, Transportation Network Modeling and Analysis, Traffic Flow Simulation, Transportation Engineering